

Smart Animal Health Assistant

Database Schema Design

This document outlines the complete database schema for the Smart Animal Health Assistant application. The system uses a MySQL relational database with five normalized tables to manage animal profiles, symptom logs, health assessments, vaccination records, and user accounts.

Database Overview

Database Name	animal_health_db
DBMS	MySQL 8.0 or higher
Character Set	UTF-8 (utf8mb4)
Storage Engine	InnoDB
Total Tables	5

Table 1: animals

Stores complete profile information for each registered animal.

Field Name	Data Type	Constraints	Description
animal_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique identifier for each animal
name	VARCHAR(100)	NOT NULL	Animal name
species	VARCHAR(50)	NOT NULL	Species (e.g., Dog, Cat, Cow)
breed	VARCHAR(100)	NULL	Breed or subspecies
age	INT	NULL	Age in years
weight	DECIMAL(6,2)	NULL	Weight in kg
animal_type	ENUM	'pet', 'farm', 'stray'	Classification of animal
owner_name	VARCHAR(100)	NULL	Owner or caretaker name
contact_number	VARCHAR(15)	NULL	Contact phone number
registration_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date of profile creation

Table 2: symptoms

Records each symptom observation with timestamp and link to animal profile.

Field Name	Data Type	Constraints	Description
symptom_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique symptom entry ID
animal_id	INT	FOREIGN KEY → animals(animal_id)	Reference to animal profile
reduced_appetite	BOOLEAN	DEFAULT FALSE	Loss of appetite observed
fever	BOOLEAN	DEFAULT FALSE	Fever detected
vomiting	BOOLEAN	DEFAULT FALSE	Vomiting observed
low_activity	BOOLEAN	DEFAULT FALSE	Lethargy or inactivity
limping	BOOLEAN	DEFAULT FALSE	Limping or difficulty walking
other_symptoms	TEXT	NULL	Additional notes/observations
recorded_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	When symptom was recorded

Table 3: health_assessments

Stores output from the rule-based inference engine: risk level, diagnosis, and recommendations.

Field Name	Data Type	Constraints	Description
assessment_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique assessment ID
symptom_id	INT	FOREIGN KEY → symptoms(symptom_id)	Reference to symptom entry
animal_id	INT	FOREIGN KEY → animals(animal_id)	Reference to animal profile
risk_level	ENUM	'Low', 'Medium', 'High'	Assessed severity level
possible_condition	VARCHAR(200)	NULL	Probable diagnosis (rule output)
recommended_action	TEXT	NULL	Action advice (Monitor/Vet/Emergency)
assessment_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	When assessment was generated

Table 4: vaccinations

Manages vaccination schedule and reminder alerts for each animal.

Field Name	Data Type	Constraints	Description
vaccination_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique vaccination record ID
animal_id	INT	FOREIGN KEY → animals(animal_id)	Reference to animal profile
vaccine_name	VARCHAR(100)	NOT NULL	Name of vaccine (e.g., Rabies, DHPP)
scheduled_date	DATE	NOT NULL	Due date for vaccination
administered_date	DATE	NULL	Actual date when given (NULL if pending)
status	ENUM	'Pending', 'Completed', 'Overdue'	Current vaccination status
notes	TEXT	NULL	Additional notes or vet instructions
created_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	When record was created

Table 5: users (Optional - for multi-user systems)

User authentication and authorization for the application (can be omitted for single-user desktop version).

Field Name	Data Type	Constraints	Description
user_id	INT	PRIMARY KEY, AUTO_INCREMENT	Unique user ID
username	VARCHAR(50)	UNIQUE, NOT NULL	Login username
password_hash	VARCHAR(255)	NOT NULL	Encrypted password (use bcrypt/SHA-256)
full_name	VARCHAR(100)	NULL	User full name
email	VARCHAR(100)	UNIQUE, NULL	User email address
role	ENUM	'Admin', 'User'	User access level
created_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp

Database Relationships

The database enforces referential integrity through foreign key constraints:

Relationship	Type	Description
animals → symptoms	One-to-Many	One animal can have multiple symptom records
animals → health_assessments	One-to-Many	One animal can have multiple assessments
symptoms → health_assessments	One-to-One	Each symptom entry generates one assessment
animals → vaccinations	One-to-Many	One animal can have multiple vaccination records
users → animals (optional)	One-to-Many	One user can manage multiple animal profiles

Sample SQL CREATE Statements

Below is a simplified example of the SQL statements required to create the database schema. In practice, you would also add indexes, CHECK constraints, and additional optimizations.

```
CREATE DATABASE animal_health_db CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;

USE animal_health_db;

-- Table 1: animals
CREATE TABLE animals (
  animal_id INT PRIMARY KEY AUTO_INCREMENT,
  name VARCHAR(100) NOT NULL,
  species VARCHAR(50) NOT NULL,
  breed VARCHAR(100),
  age INT,
  weight DECIMAL(6,2),
  animal_type ENUM('pet', 'farm', 'stray') NOT NULL,
  owner_name VARCHAR(100),
  contact_number VARCHAR(15),
  registration_date DATETIME DEFAULT CURRENT_TIMESTAMP
);

-- Table 2: symptoms
CREATE TABLE symptoms (
  symptom_id INT PRIMARY KEY AUTO_INCREMENT,
  animal_id INT NOT NULL,
  reduced_appetite BOOLEAN DEFAULT FALSE,
  fever BOOLEAN DEFAULT FALSE,
  vomiting BOOLEAN DEFAULT FALSE,
  low_activity BOOLEAN DEFAULT FALSE,
  limping BOOLEAN DEFAULT FALSE,
  other_symptoms TEXT,
  recorded_date DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (animal_id) REFERENCES animals(animal_id) ON DELETE CASCADE
);

-- Table 3: health_assessments
CREATE TABLE health_assessments (
  assessment_id INT PRIMARY KEY AUTO_INCREMENT,
  symptom_id INT NOT NULL,
  animal_id INT NOT NULL,
  risk_level ENUM('Low', 'Medium', 'High') NOT NULL,
  possible_condition VARCHAR(200),
  recommended_action TEXT,
  assessment_date DATETIME DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (symptom_id) REFERENCES symptoms(symptom_id) ON DELETE CASCADE,
  FOREIGN KEY (animal_id) REFERENCES animals(animal_id) ON DELETE CASCADE
);

-- Table 4: vaccinations
```

```
CREATE TABLE vaccinations (  
  vaccination_id INT PRIMARY KEY AUTO_INCREMENT,  
  animal_id INT NOT NULL,  
  vaccine_name VARCHAR(100) NOT NULL,  
  scheduled_date DATE NOT NULL,  
  administered_date DATE,  
  status ENUM('Pending', 'Completed', 'Overdue') DEFAULT 'Pending',  
  notes TEXT,  
  created_date DATETIME DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (animal_id) REFERENCES animals(animal_id) ON DELETE CASCADE  
);
```

-- Table 5: users (optional)

```
CREATE TABLE users (  
  user_id INT PRIMARY KEY AUTO_INCREMENT,  
  username VARCHAR(50) UNIQUE NOT NULL,  
  password_hash VARCHAR(255) NOT NULL,  
  full_name VARCHAR(100),  
  email VARCHAR(100) UNIQUE,  
  role ENUM('Admin', 'User') DEFAULT 'User',  
  created_date DATETIME DEFAULT CURRENT_TIMESTAMP  
);
```